

Project Design Phase
Problem – Solution Fit Template

Date	19 July 2026
Team ID	
Project Name	AI-Driven Emotion Detection & Personalized Learning Support Platform
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

1. CUSTOMER SEGMENT(S) CS Who is your customer? i.e. working parents of 0-5 yrs kids • College students • University learners • Self-learning students • Beginners in AI and Programming • Students preparing for exams and interviews Define CS, fit into CC	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices. • Difficulty understanding concepts • Lack of personalized guidance • Low motivation during learning • Limited access to mentors • Time management issues	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital learning • YouTube tutorials • Google Search • ChatGPT and Gemini • Online learning platforms • Discussion forums Explore AS, differentiate
2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customer? There could be more than one, explore different sides. Students need a platform that understands their learning emotions and provides personalized study guidance to improve learning effectiveness. Focus on J&P, tap into BE, understand RC	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? i.e. customers have to do it because of the change in regulations • No emotion-aware learning support • Generic learning resources • Lack of instant feedback • Difficulty identifying learning challenges	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? i.e. directly related: find the right note, paper, notes, calculate usage and benefits, indirectly associated: customers spend less time on understanding work (i.e. time-gain) Students often feel confused, frustrated, or bored while studying and switch between multiple resources without finding suitable guidance. Focus on J&P, tap into BE, understand RC
3. TRIGGERS TR What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. • Difficult concepts • Assignment deadlines • Examination pressure • Low confidence • Lack of motivation 4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? i.e. best, separate + confident, in control - use it in your communication strategy & design. Before: Confident, Frustrated, Bored After: Confused, Motivated, Calm, Focused Identify strong TR & EM	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. Develop an AI-driven Emotion Detection and Personalized Learning Support Platform that detects students' emotions from text input and provides personalized AI-generated learning guidance using BERT, BiLSTM, rule-based enhancement, and Gemini API.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 • Web Application • GitHub Repository • College Projects • Educational Institutions 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. Teachers, seniors, friends, textbooks, and YouTube are used for guidance, but they do not provide personalized emotion-based learning support. Extract online & offline CH of BE

References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>